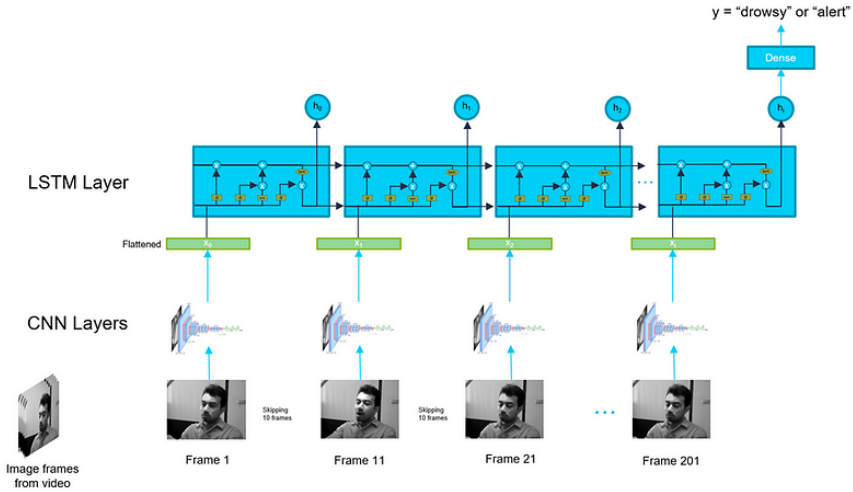
Recurrent Neural Networks

Naeemullah Khan
naeemullah.khan@kaust.edu.sa

جامعة الملك عبدالله
للعلوم والتقنية
King Abdullah University of
Science and Technology

KAUST Academy
King Abdullah University of Science and Technology

June 29, 2026



Source: junfengzhang.com

Table of Contents

Motivation

Why this topic matters:

- Many real-world signals are sequential: text, speech, video, and time series.
- Standard CNNs and MLPs expect fixed-size inputs and ignore order.
- RNNs process variable-length sequences while sharing weights across time.
- They are the foundation for sequence-to-sequence learning and the attention mechanism.
- *"Understanding RNNs and their limits is the gateway to modern Transformers."*

⇒ Learning Outcomes

By the end of this session, you will be able to:

- Describe how an RNN processes sequential data with a recurrent hidden state.
- Explain the advantages and limitations of vanilla RNNs (e.g., vanishing/exploding gradients).
- Understand how gated units (LSTM, GRU) mitigate long-term dependency problems.
- Apply RNNs to vision-language tasks such as image captioning.
- Describe the sequence-to-sequence framework and the motivation for attention.

Advanced Computer Vision: **Recurrent Neural Networks (RNN)**

Recurrent Neural Networks (RNN)

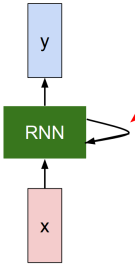
What are RNNs?

- Neural networks designed for sequence data.
- Maintain a hidden state h_t that carries information from previous time steps.
- Recurrent architecture: $h_t = f(h_{t-1}, x_t)$

Use Cases:

- Language modeling
- Time series prediction
- Sequence classification

Recurrent Neural Networks (RNN) (cont.)



Key idea: RNNs have an "internal state" that is updated as a sequence is processed

$$h_t = f_W(h_{t-1}, x_t)$$

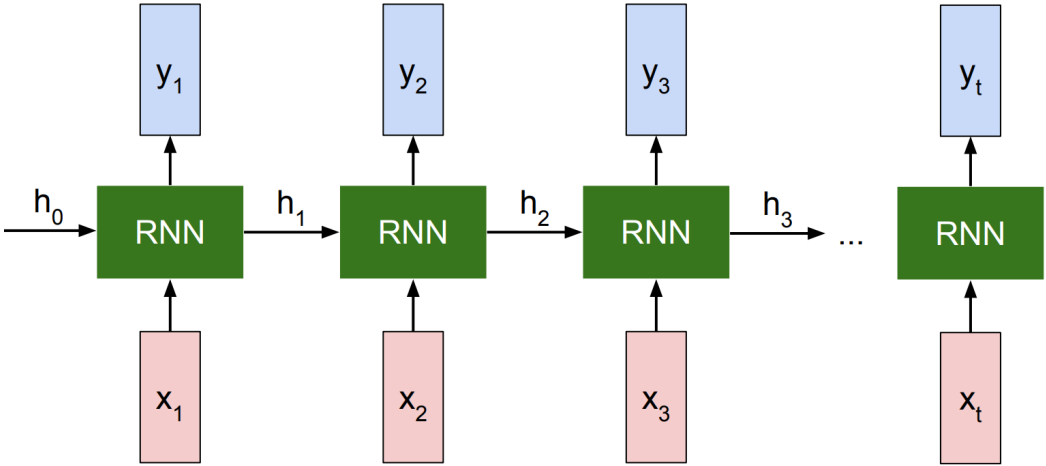
new state

some function with parameters W

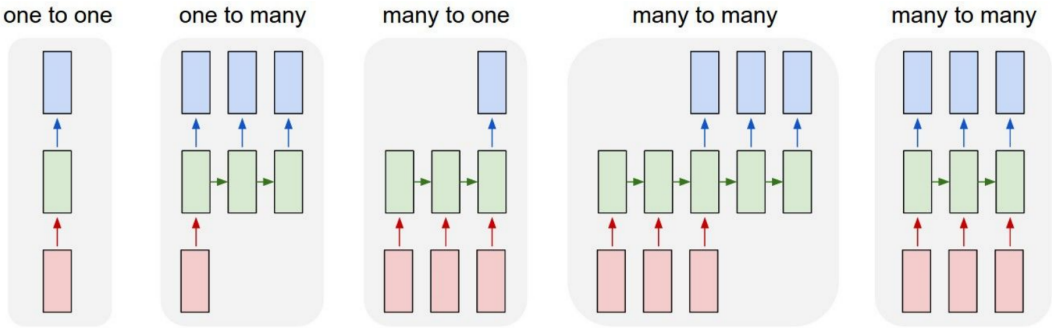
old state

input vector at some time step

Recurrent Neural Networks (RNN) (cont.)



Recurrent Neural Networks (RNN) (cont.)



Recurrent Neural Networks (RNN) (cont.)

RNN Advantages:

- Can process any length input
- Computation for step t can (in theory) use information from many steps back
- Model size doesn't increase for longer input
- Same weights applied on every timestep, so there is symmetry in how inputs are processed.

Recurrent Neural Networks (RNN) (cont.)

RNN Disadvantages:

- Recurrent computation is slow
- In practice, difficult to access information from many steps back

Limitations of RNNs:

- Vanishing/exploding gradients in long sequences.
- Sequential computation — slow to train.
- Difficulty in capturing long-term dependencies.
- Cannot parallelize easily due to recursive nature.

Recurrent Neural Networks: **Gated Units: LSTM & GRU**

The Problem: Long-Term Dependencies

- Vanilla RNNs repeatedly multiply by the same weight matrix at every step.
- Gradients tend to **vanish** (shrink to zero) or **explode** over long sequences.
- As a result, plain RNNs struggle to connect information across many time steps.
- **Idea:** add gating mechanisms that let the network *choose* what to keep, forget, and output.

Long Short-Term Memory (LSTM)

- Maintains a separate **cell state** c_t that acts as a memory "conveyor belt".
- Three gates control the flow of information:
  - **Forget gate** f_t: what to erase from memory.
  - **Input gate** i_t: what new information to store.
  - **Output gate** o_t: what to expose as the hidden state.
- Additive cell updates create a gradient "highway", easing long-range learning.

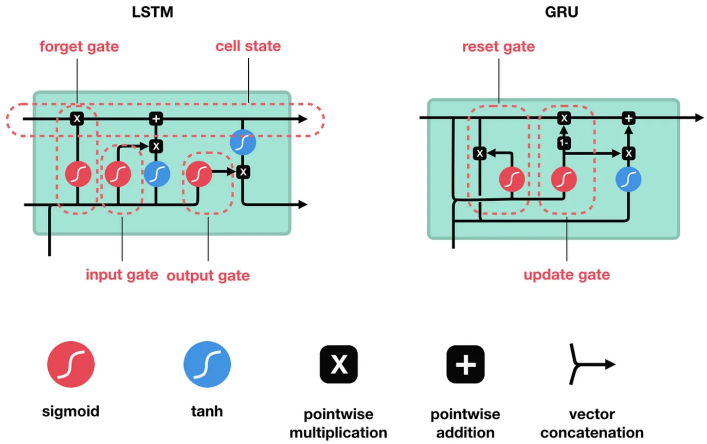
$$
\begin{aligned}
f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) \\
i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) \\
o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) \\
\tilde{c}_t &= \tanh(W_c[h_{t-1}, x_t] + b_c) \\
c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \\
h_t &= o_t \odot \tanh(c_t)
\end{aligned}
$$

Gated Recurrent Unit (GRU)

- A lighter alternative to the LSTM with **two** gates and no separate cell state.
- **Reset gate** r_t: how much past state to ignore.
- **Update gate** z_t: how much to blend old and new state.
- Fewer parameters $\Rightarrow$ faster to train, often with comparable accuracy.

$$z_t = \sigma(W_z[h_{t-1}, x_t])$$
$$r_t = \sigma(W_r[h_{t-1}, x_t])$$
$$\tilde{h}_t = \tanh(W[r_t \odot h_{t-1}, x_t])$$
$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

LSTM vs GRU



Source: Illustrated Guide to LSTM's and GRU's: A step by step explanation

LSTM vs GRU

Aspect	LSTM	GRU
Gates	3 (input, forget, output)	2 (reset, update)
Separate cell state	Yes	No
Parameters	More	Fewer
Training speed	Slower	Faster
Typical use	Long, complex sequences	Smaller data / faster models

- Both dramatically outperform vanilla RNNs on long-range dependencies.
- The choice is empirical — GRU is a strong, cheaper default; LSTM can edge ahead on hard tasks.

Advanced Computer Vision: **Image Captioning**

Image Captioning

- A computer Vision task in which we generate captions for images

A cat sitting on a suitcase on the floor

A cat is sitting on a tree branch

A dog is running in the grass with a frisbee

A white teddy bear sitting in the grass

Two people walking on the beach with surfboards

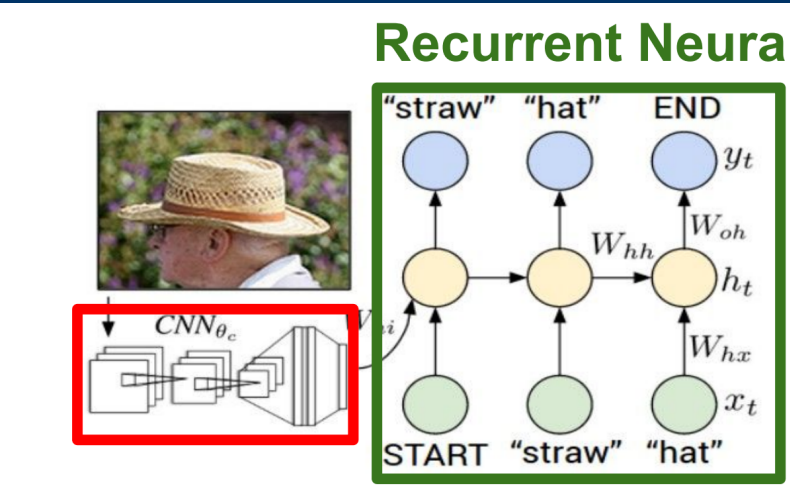
A tennis player in action on the court

Two giraffes standing in a grassy field

A man riding a dirt bike on a dirt track

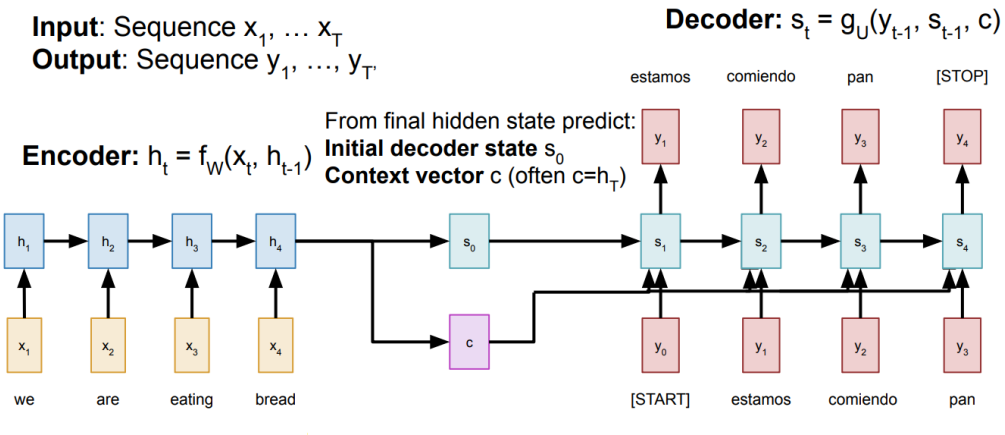
Image Captioning (cont.)

- The task is to generate a natural language description of the content of an image
- It combines techniques from computer vision and natural language processing
- The process typically involves:
  - Extracting features from the image using a convolutional neural network (CNN)
  - Using these features to generate a sequence of words that describe the image, often with a recurrent neural network (RNN) or transformer model


Recurrent Neural Network
"straw"
"hat"
END
y_t
W_{oh}
W_{hh}
h_t
W_{hx}
x_t
START
"straw"
"hat"
CNN_{θ_c}
Convolutional Neural Network

Advanced Computer Vision: **Sequence-to-Sequence with RNNs**

Sequence-to-Sequence with RNNs



Sutskever et al, "Sequence to sequence learning with neural networks

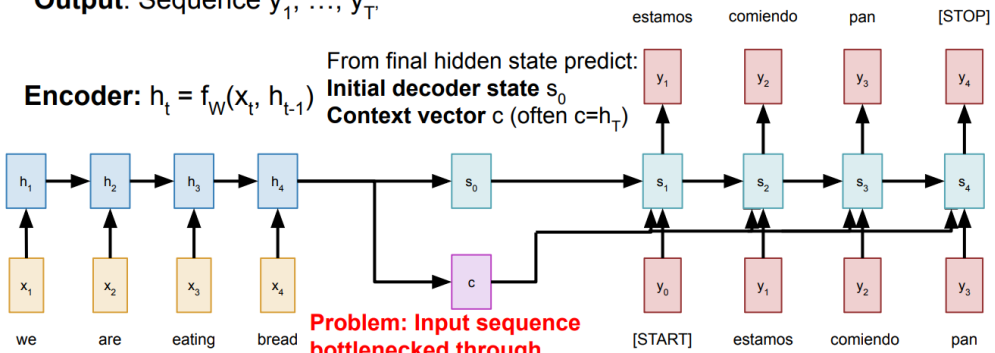
Sequence-to-Sequence with RNNs (cont.)

Input: Sequence $x_1, \ldots x_T$
Output: Sequence $y_1, \ldots, y_{T'}$

Encoder: $h_t = f_W(x_t, h_{t-1})$

Decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c)$

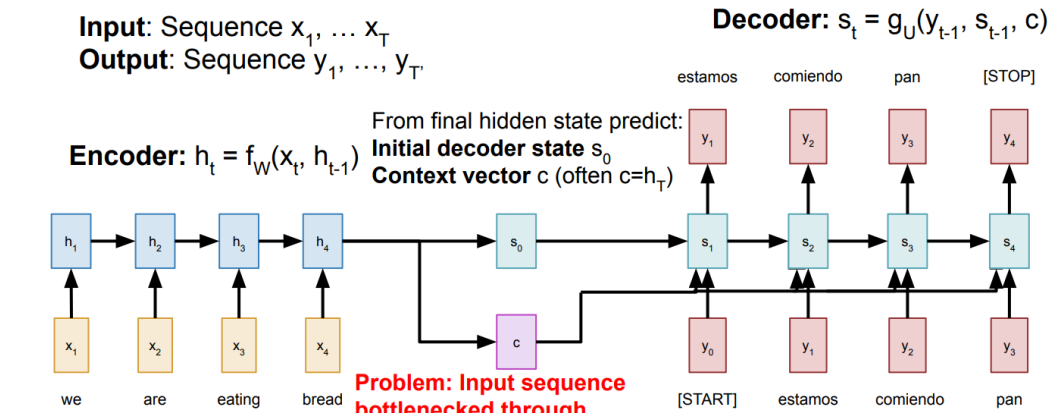
From final hidden state predict:
Initial decoder state s_0
Context vector c (often $c=h_T$)



Problem: Input sequence bottlenecked through fixed-sized vector. What if T=1000?

Sutskever et al, "Sequence to sequence learning with neural networks", NeurIPS 2014

Sequence-to-Sequence with RNNs (cont.)



Sutskever et al, "Sequence to sequence learning with neural networks", NeurIPS 2014

Introduction to Attention

- **Key idea:** "Don't process everything — just focus on the most relevant parts."
- Attention helps models decide where to look in a sequence.

Equation (simplified):

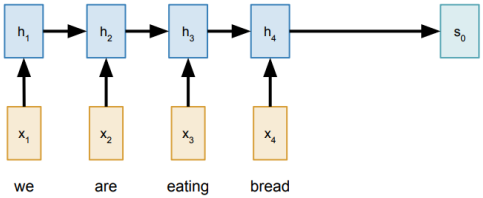
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

- Q = Queries
- K = Keys
- V = Values

Sequence-to-Sequence with RNNs and Attention

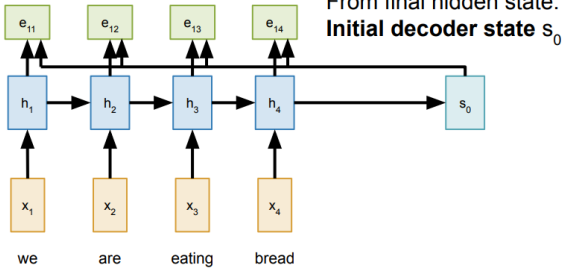
Input: Sequence $x_1, \ldots x_T$
Output: Sequence $y_1, \ldots, y_{T'}$

Encoder: $h_t = f_W(x_t, h_{t-1})$ From final hidden state: **Initial decoder state** s_0



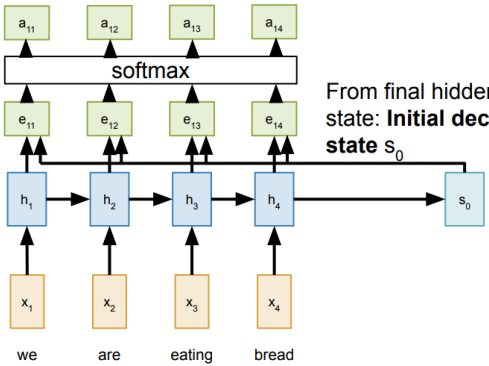
Sequence-to-Sequence with RNNs and Attention (cont.)

Compute (scalar) **alignment scores**
$e_{t,i} = f_{att}(s_{t-1}, h_i)$ (f_{att} is an MLP)



From final hidden state:
Initial decoder state s_0

Sequence-to-Sequence with RNNs and Attention (cont.)



From final hidden state: **Initial decoder state** s_0

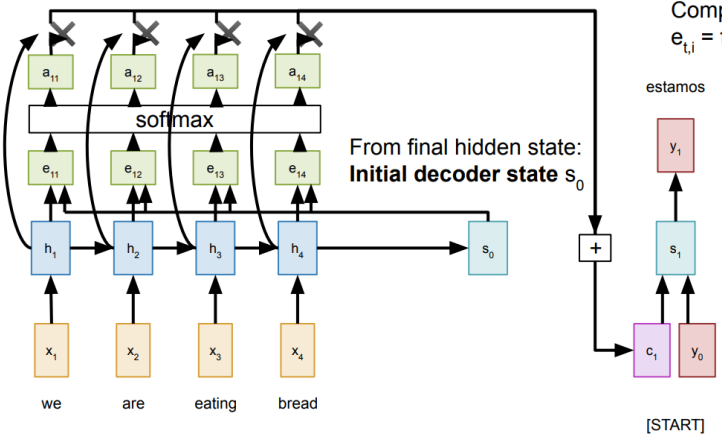
Compute (scalar) **alignment scores**

$e_{t,i} = f_{att}(s_{t-1}, h_i)$ (f_{att} is an MLP)

Normalize alignment scores to get **attention weights**

$0 < a_{t,i} < 1 \quad \sum_i a_{t,i} = 1$

Sequence-to-Sequence with RNNs and Attention (cont.)



Compute (scalar) **alignment scores**

$e_{t,i} = f_{att}(s_{t-1}, h_i)$ (f_{att} is an MLP)

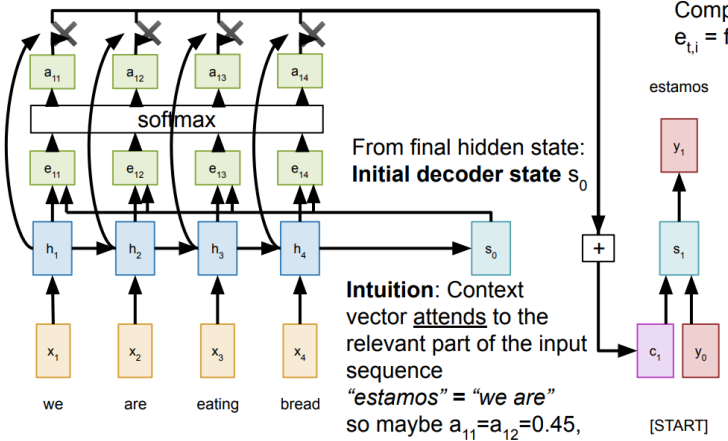
Normalize alignment scores to get **attention weights**

$0 < a_{t,i} < 1 \quad \sum_i a_{t,i} = 1$

Compute context vector as linear combination of hidden states

$c_t = \sum_i a_{t,i} h_i$

Sequence-to-Sequence with RNNs and Attention (cont.)



Compute (scalar) **alignment scores**

$e_{t,i} = f_{att}(s_{t-1}, h_i)$ (f_{att} is an MLP)

Normalize alignment scores to get **attention weights**

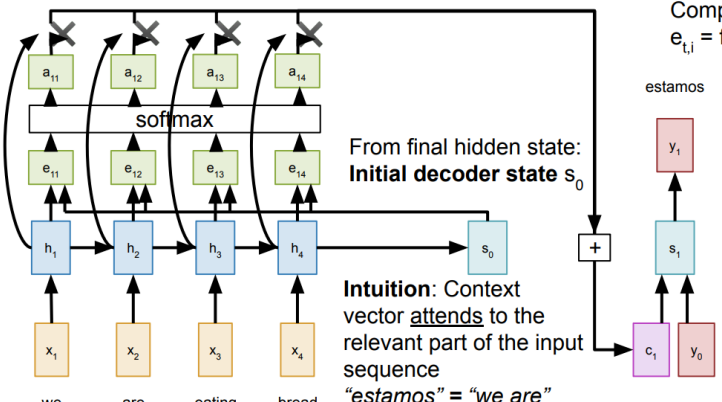
$0 < a_{t,i} < 1 \quad \sum_i a_{t,i} = 1$

Compute context vector as linear combination of hidden states

$c_t = \sum_i a_{t,i} h_i$

Use context vector in decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c_t)$

Sequence-to-Sequence with RNNs and Attention (cont.)



Compute (scalar) **alignment scores**

$e_{t,i} = f_{att}(s_{t-1}, h_i)$ (f_{att} is an MLP)

Normalize alignment scores to get **attention weights**

$0 < a_{t,i} < 1 \quad \sum_i a_{t,i} = 1$

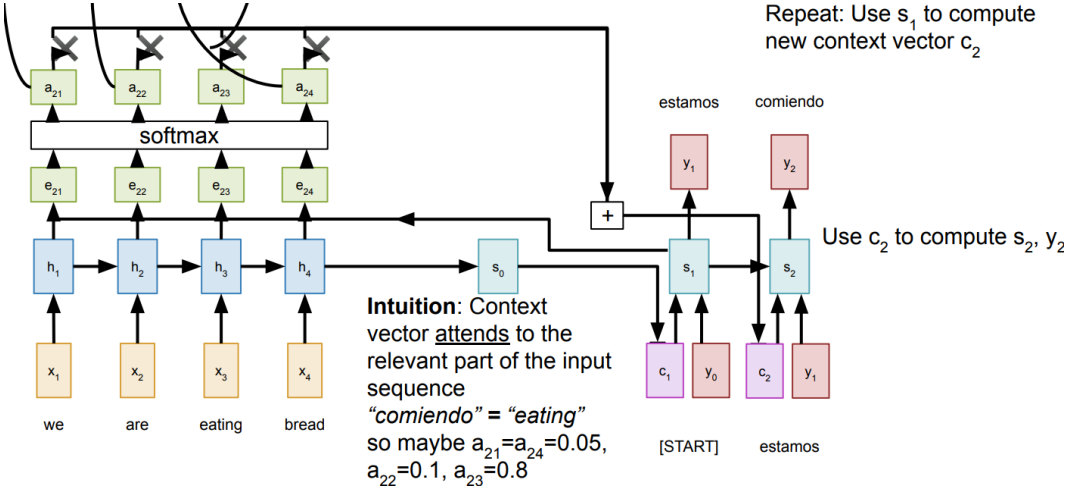
Compute context vector as linear combination of hidden states

$c_t = \sum_i a_{t,i} h_i$

Use context vector in decoder: $s_t = g_U(y_{t-1}, s_{t-1}, c_t)$

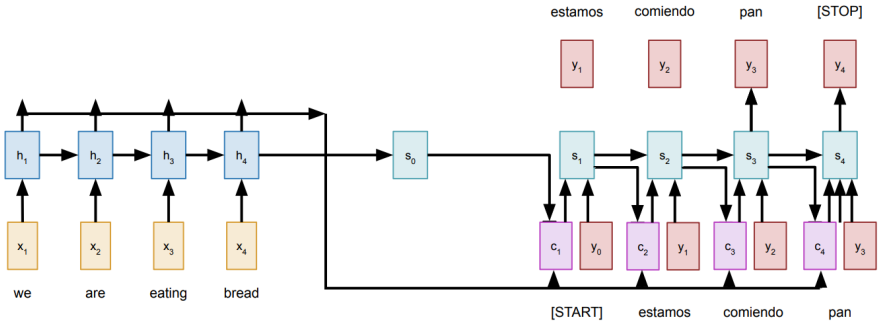
This is all differentiable! No supervision on attention weights – backprop through everything

Sequence-to-Sequence with RNNs and Attention (cont.)



Sequence-to-Sequence with RNNs and Attention (cont.)

- Use a different context vector at each timestep of the decoder.
- The input sequence is not bottlenecked through a single vector.
- At each timestep of the decoder, the context vector "looks at" different parts of the input sequence.



Sequence-to-Sequence with RNNs and Attention (cont.)

Example: English to French translation

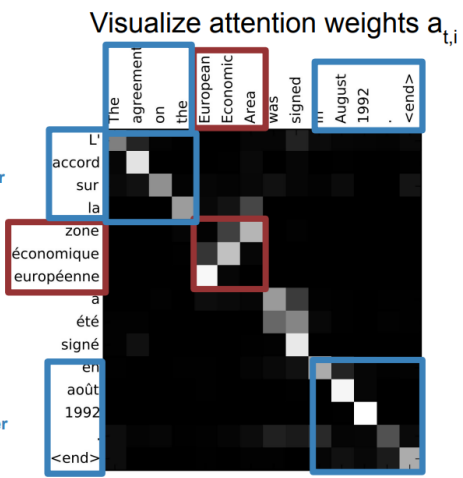
Input: "**The agreement on the European Economic Area** was signed **in August 1992**."

Output: "**L'accord sur la zone économique européenne** a été signé **en août 1992**."

Diagonal attention means words correspond in order

Attention figures out different word orders

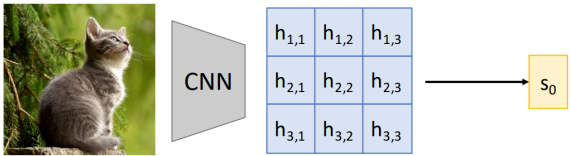
Diagonal attention means words correspond in order



Bahdanau et al, "Neural machine translation by jointly learning to align and translate", ICLR 2015

Advanced Computer Vision: **Image Captioning with RNNs and Attention**

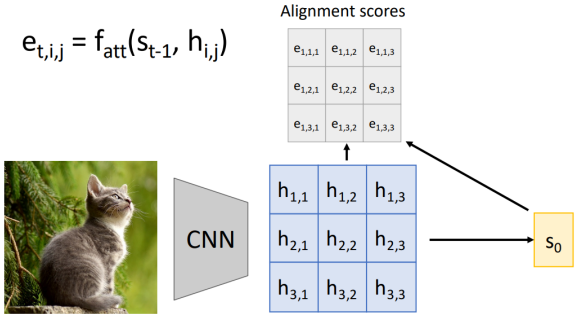
Image Captioning with RNNs and Attention



Use a CNN to compute a grid of features for an image

Image Captioning with RNNs and Attention

$e_{t,i,j} = f_{att}(s_{t-1}, h_{i,j})$

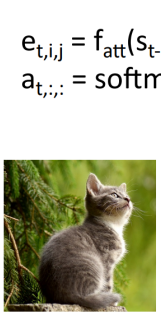


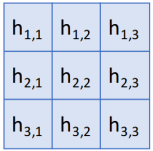
Use a CNN to compute a grid of features for an image

Image Captioning with RNNs and Attention

$e_{t,i,j} = f_{att}(s_{t-1}, h_{i,j})$

$a_{t,:,:} = \text{softmax}(e_{t,:,:})$





Use a CNN to compute a grid of features for an image

Image Captioning with RNNs and Attention

$e_{t,i,j} = f_{att}(s_{t-1}, h_{i,j})$

$a_{t,:,:} = \text{softmax}(e_{t,:,:})$

$c_t = \sum_{i,j} a_{t,i,j} h_{i,j}$

CNN

Alignment scores

$e_{1,1,1}$	$e_{1,1,2}$	$e_{1,1,3}$
$e_{1,2,1}$	$e_{1,2,2}$	$e_{1,2,3}$
$e_{1,3,1}$	$e_{1,3,2}$	$e_{1,3,3}$

softmax

Attention weights

$a_{1,1,1}$	$a_{1,1,2}$	$a_{1,1,3}$
$a_{1,2,1}$	$a_{1,2,2}$	$a_{1,2,3}$
$a_{1,3,1}$	$a_{1,3,2}$	$a_{1,3,3}$

$h_{1,1}$	$h_{1,2}$	$h_{1,3}$
$h_{2,1}$	$h_{2,2}$	$h_{2,3}$
$h_{3,1}$	$h_{3,2}$	$h_{3,3}$

s_0

c_1

Use a CNN to compute a grid of features for an image

Image Captioning with RNNs and Attention

$e_{t,i,j} = f_{att}(s_{t-1}, h_{i,j})$

$a_{t,:,:} = \text{softmax}(e_{t,:,:})$

$c_t = \sum_{i,j} a_{t,i,j} h_{i,j}$

CNN

Use a CNN to compute a grid of features for an image

Alignment scores

$e_{1,1,1}$	$e_{1,1,2}$	$e_{1,1,3}$
$e_{1,2,1}$	$e_{1,2,2}$	$e_{1,2,3}$
$e_{1,3,1}$	$e_{1,3,2}$	$e_{1,3,3}$

softmax

Attention weights

$a_{1,1,1}$	$a_{1,1,2}$	$a_{1,1,3}$
$a_{1,2,1}$	$a_{1,2,2}$	$a_{1,2,3}$
$a_{1,3,1}$	$a_{1,3,2}$	$a_{1,3,3}$

$h_{1,1}$	$h_{1,2}$	$h_{1,3}$
$h_{2,1}$	$h_{2,2}$	$h_{2,3}$
$h_{3,1}$	$h_{3,2}$	$h_{3,3}$

s_0 s_1 y_1 c_1 y_0

cat

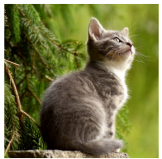
[START]

Image Captioning with RNNs and Attention

$e_{t,i,j} = f_{att}(s_{t-1}, h_{i,j})$

$a_{t,:,:} = \text{softmax}(e_{t,:,:})$

$c_t = \sum_{i,j} a_{t,i,j} h_{i,j}$



CNN

$h_{1,1}$ $h_{1,2}$ $h_{1,3}$
$h_{2,1}$ $h_{2,2}$ $h_{2,3}$
$h_{3,1}$ $h_{3,2}$ $h_{3,3}$

s_0 s_1

c_1 y_0 y_1

cat

[START]

Use a CNN to compute a grid of features for an image

Image Captioning with RNNs and Attention

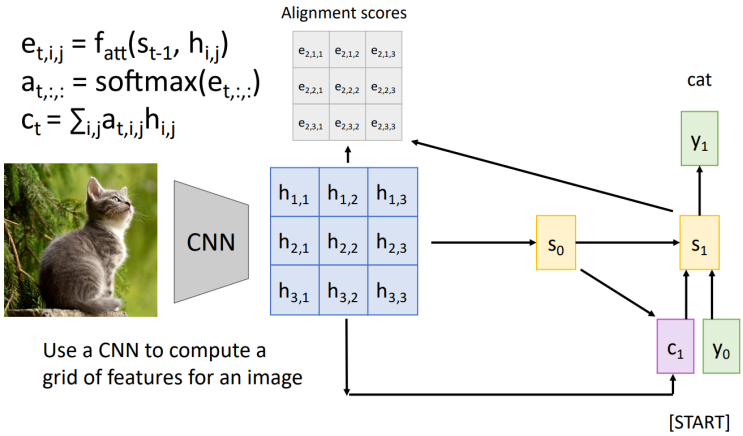


Image Captioning with RNNs and Attention

$e_{t,i,j} = f_{att}(s_{t-1}, h_{i,j})$

$a_{t,:,:} = \text{softmax}(e_{t,:,:})$

$c_t = \sum_{i,j} a_{t,i,j} h_{i,j}$

CNN

Use a CNN to compute a grid of features for an image

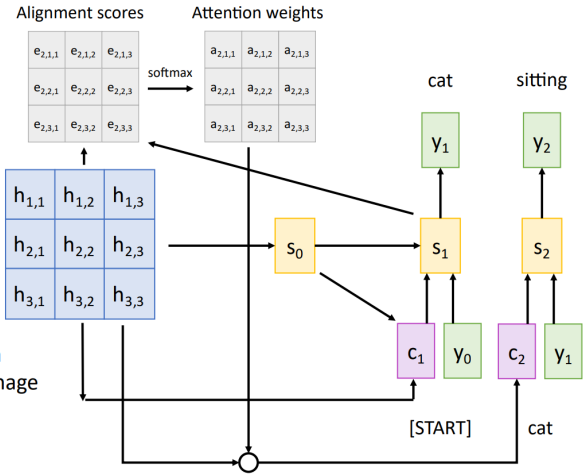


Image Captioning with RNNs and Attention

$e_{t,i,j} = f_{att}(s_{t-1}, h_{i,j})$
$a_{t,:,:} = \text{softmax}(e_{t,:,:})$
$c_t = \sum_{i,j} a_{t,i,j} h_{i,j}$

Each timestep of decoder uses a different context vector that looks at different parts of the input image

CNN

$h_{1,1}$ $h_{1,2}$ $h_{1,3}$
$h_{2,1}$ $h_{2,2}$ $h_{2,3}$
$h_{3,1}$ $h_{3,2}$ $h_{3,3}$

Use a CNN to compute a grid of features for an image

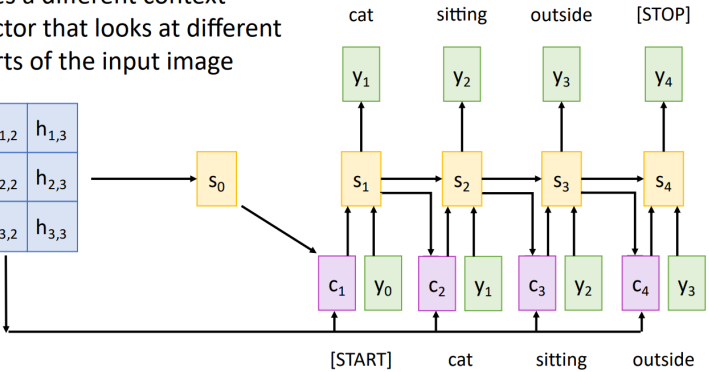


Image Captioning with RNNs and Attention

A dog is standing on a hardwood floor.

A stop sign is on a road with a mountain in the background.

A group of people sitting on a boat in the water.

A giraffe standing in a forest with trees in the background.

Recurrent Neural Networks: **Limitations and Future Directions**

Limitations of RNNs

- **Sequential computation**: cannot be parallelized across time steps (slow to train).
- **Long-range dependencies**: even LSTMs/GRUs struggle on very long sequences.
- **Limited memory**: the entire context is squeezed into a fixed-size hidden state.
- **Bottleneck in seq2seq**: a single context vector limits the decoder.

Future Directions

- **Attention**: let the decoder look at all encoder states (Bahdanau et al.; next topic).
- **Transformers**: drop recurrence entirely for full parallelism (Vaswani et al.).
- **Efficient sequence models**: e.g., state-space models (S4, Mamba).

Recurrent Neural Networks: **References**

References

[1] Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780.

[2] Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. *EMNLP*. https://arxiv.org/abs/1406.1078

[3] Sutskever, I., Vinyals, O., & Le, Q. V. (2014). Sequence to Sequence Learning with Neural Networks. *Advances in Neural Information Processing Systems*, 27. https://arxiv.org/abs/1409.3215

[4] Bahdanau, D., Cho, K., & Bengio, Y. (2015). Neural Machine Translation by Jointly Learning to Align and Translate. *International Conference on Learning Representations*. https://arxiv.org/abs/1409.0473

References (cont.)

[5] Xu, K., Ba, J., Kiros, R., Cho, K., Courville, A., Salakhutdinov, R., Zemel, R., & Bengio, Y. (2015). Show, Attend and Tell: Neural Image Caption Generation with Visual Attention. *International Conference on Machine Learning*. https://arxiv.org/abs/1502.03044

[6] Fei-Fei Li, Johnson, J., & Yeung, S. Stanford CS231n: Deep Learning for Computer Vision. http://cs231n.stanford.edu/